PROJECT DOCUMENTATION

SAP P2P SIM

Procure-to-Pay Lifecycle & Financial Integration Report

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Course: SAP ABAP

PROJECT REF: Procure-to-Pay (P2P) - Full Purchasing Cycle (SAP MM)

live demo: https://sap-p2p.vercel.app/dashboard

1. Problem Statement

In modern corporate environments, procurement is often hindered by fragmented processes, lack of data transparency, and poor financial oversight. Traditional manual systems fail to enforce critical business rules, leading to "maverick buying," duplicate payments, and budget overruns. Educational simulations often simplify these processes too much, failing to teach students the rigorous internal controls—such as Master Data Governance and the Three-Way Match—required in real-world Enterprise Resource Planning (ERP) systems like SAP.

2. Solution Overview

The SAP P2P Management System is a comprehensive digital solution that simulates the entire lifecycle of corporate procurement. By integrating purchasing, inventory management, and accounts payable, the system provides a "Single Source of Truth." It enforces a strict, status-based workflow that ensures financial accountability and operational efficiency, mirroring the SAP MM (Materials Management) and FI (Financial Accounting) modules.

3. Core Features

3.1 End-to-End P2P Lifecycle

The system automates and enforces the following five critical stages:

Document Type	Standard T-Code	Action Description	Financial Impact
Requisition (PR)	`ME51N`	Internal request created by a cost center.	None (Commitment only)
Purchase Order (PO)	`ME21N`	Legal contract issued to Vendor Master.	Encumbrance created
Goods Receipt (GR)	`MIGO`	Logistics confirmation of delivery.	Stock/Expense update
Invoice Receipt (IR)	`MIRO`	Verification of bill against receipt.	Liability (AP) recognized
Payment (FI)	`F-53`	Settlement via UPI/Bank/Card.	Cash outflow

3.2 Master Data Governance

The system eliminates manual data errors by utilizing centralized Material and Vendor Master records, ensuring that every transaction uses pre-verified pricing and supplier details.

3.3 Real-time Dashboard & Analytics

Management receives instant visibility through a dynamic dashboard tracking total spend, pending liabilities, and supplier performance metrics.

4. Tech Stack

Frontend: Next.js 15 (App Router Architecture)

Logic: React.js Hooks (Client-side state management)

Styling: Tailwind CSS (Utility-first CSS for professional UI)

Persistence: Browser LocalStorage (No-DB persistent data)

Icons: Lucide-React (Enterprise-standard iconography)

5. System Screenshots

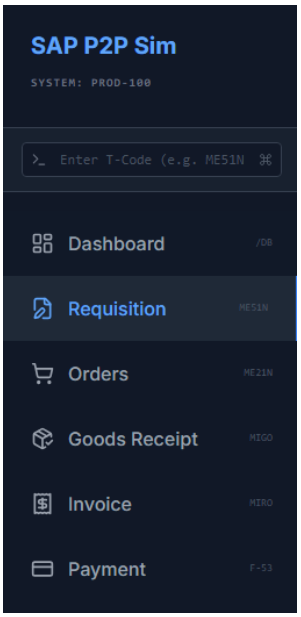

Figure 1: Sidebar Navigation with T-Code Command Interface

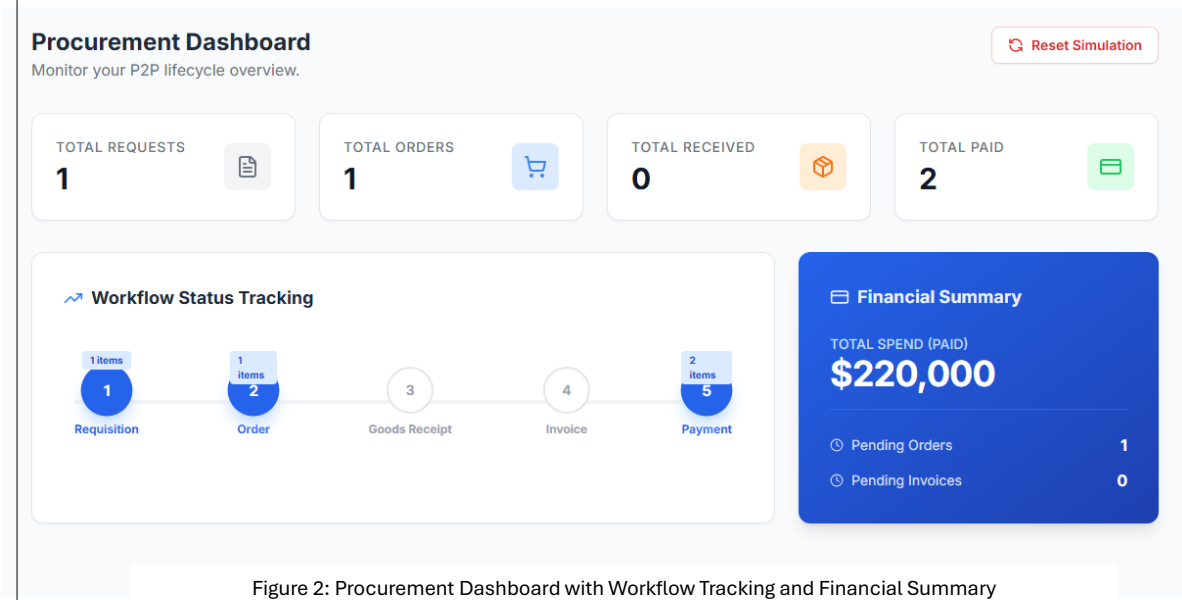

Figure 2: Procurement Dashboard with Workflow Tracking and Financial Summary

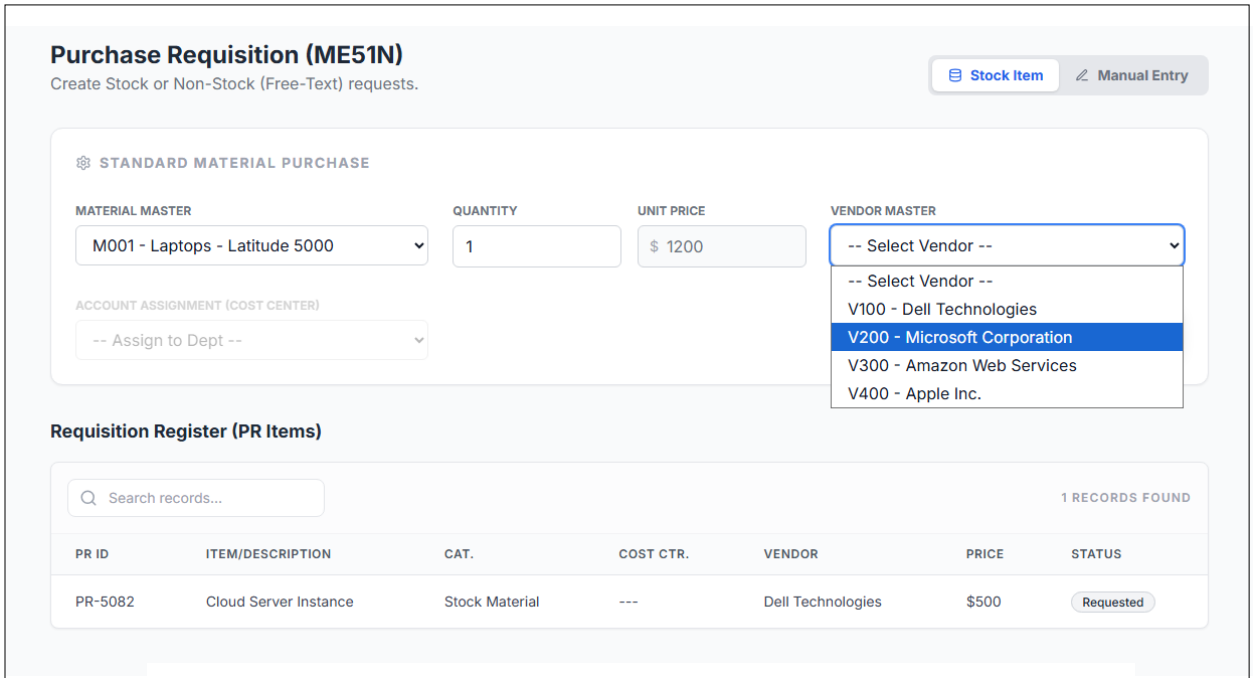

Figure 3: Purchase Requisition (ME51N) Interface with Material and Vendor Selection

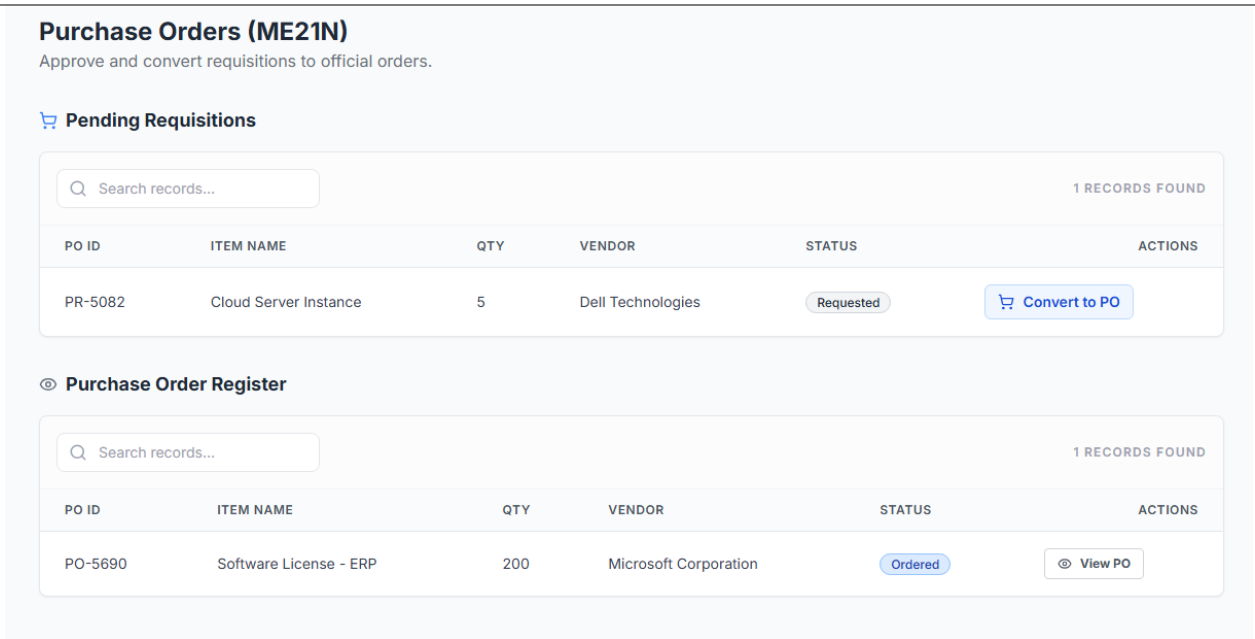

Figure 4: Purchase Orders (ME21N) Interface with Requisition Conversion and Order Register

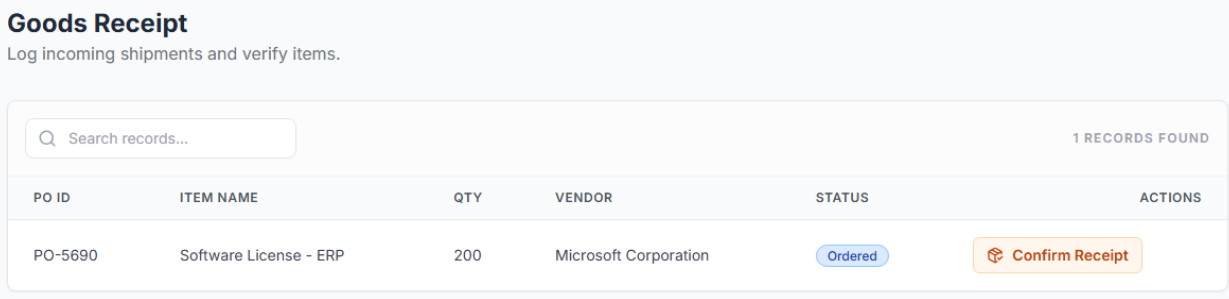

Figure 5: Goods Receipt Interface with Order Verification and Receipt Confirmation

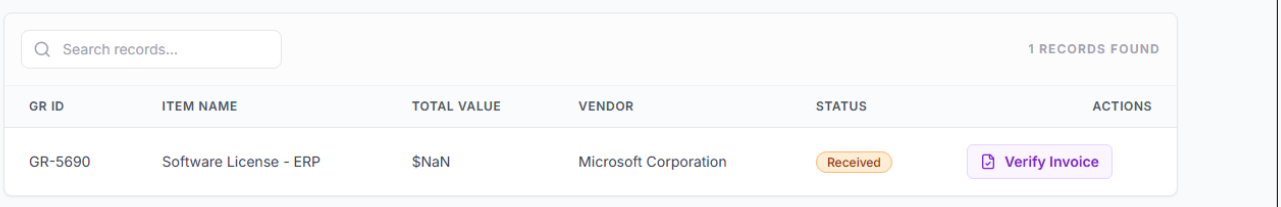

Figure 6: Invoice Verification Interface with Goods Receipt Matching and Invoice Processing

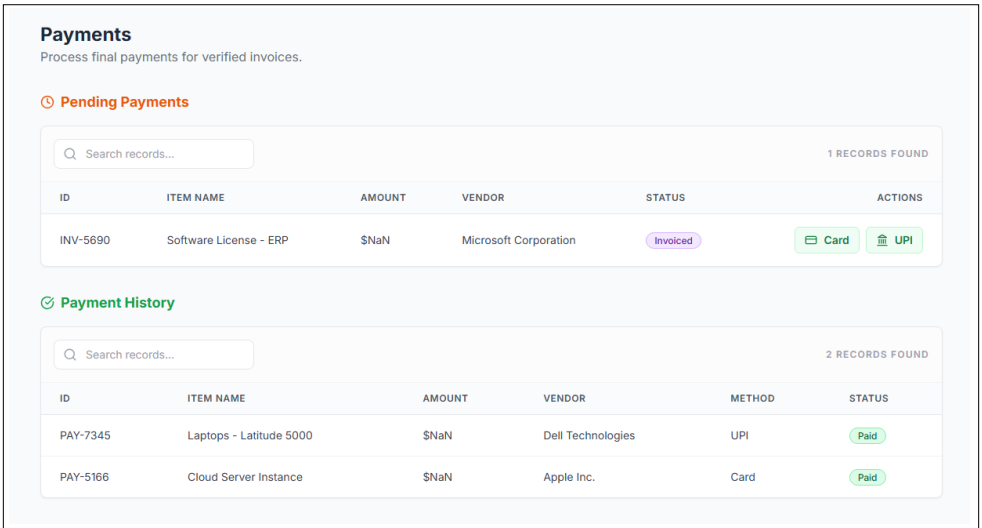

Figure 7: Payment Interface with Pending Payments and Payment History Tracking

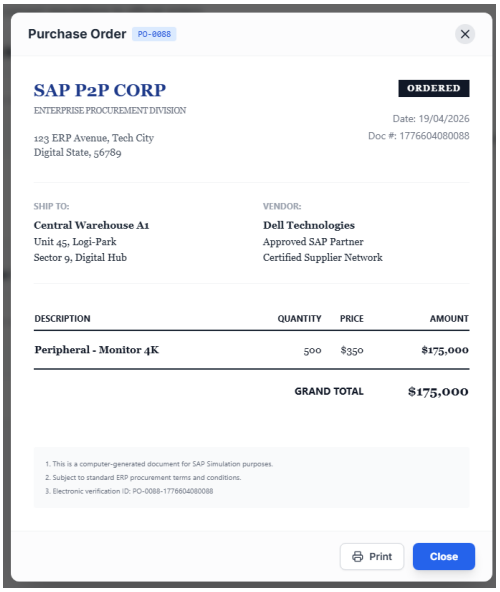

Figure 8: Purchase Order Document Interface with Vendor Details and Financial Summary

6. Unique Points:

- Implements SAP-like T-Code navigation (ME51N, MIGO) for fast access
- Uses Cost Center logic for non-stock items (Account Assignment ‘K’)
- Enforces Three-Way Match to ensure accurate and controlled payments
- Simulates real ERP document flow and enterprise-level logic

7. Future Improvements:

- Add approval workflows for high-value requisitions
- Integrate advanced reporting with charts and analytics
- Support multi-currency for global procurement
- Migrate to cloud database (Supabase/PostgreSQL) for scalability
- Enable multi-user collaboration and role-based access control